

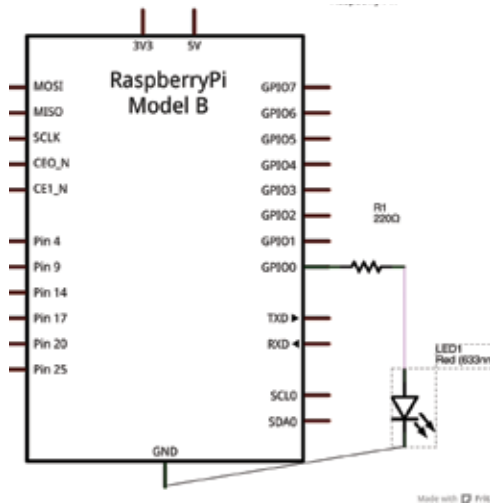
These commands are instructions for the assembler itself so that it translates code correctly. The first line of the code tells the assembler where to place the code itself. The downloaded template you have instructs the code in the section “.init” to be at the beginning of the output. We give this instruction so that the our code is first to be executed, otherwise the default setting would cause the code to be executed in an ascending alphabetical order. The “.section” command informs the assembler which section the code belongs in until we give it a new “.section” location or the code comes to an end. The last two lines are only to stop a warning message and don’t require that much attention.

Once the instructions to the assembler have been coded we now move on to the real programming. All new instructions in assembly code need to be in a new line so that the processor moves on progressively from line to line. The next line of code is:

ldr r0,=0x20200000

This is the first instruction to the processor. This line tells the processor to store the number 0x20200000 into the

register r0. In this case, a register is a tiny allocated memory component in the processor where the processor can store the numbers it requires in its active program. There are many registers that have different meanings and purposes which we won’t be discussing now. We are using the simplest of these registers called General Purpose, which range from r0 to r12. They can be used for any calculations required of the processor. We can use any General Purpose register but for sake of simplicity we are using the r0 here. As long as we maintain consistency in our programming it doesn’t matter which register we use.



Assembly coding is the way to control the hardware reactions of any processor board.